

Chapter 8B

Respiratory, Tuberculosis, Influenza, and Central Nervous System Infections

Medvale Internal Medicine - Part VIII: Infectious Diseases

This chapter is an original Medvale synthesis. The uploaded source section was used as a coverage signal for pneumonia, hospital/ventilator-associated pneumonia, lung abscess, tuberculosis, influenza, and meningitis; clinical details were cross-checked against current guideline sources listed at the end.

Why this matters clinically

Respiratory and central nervous system infections are common, time-sensitive, and pattern-rich. Pneumonia can range from self-limited outpatient illness to septic shock and acute respiratory failure. Tuberculosis is both an individual disease and a public-health diagnosis because airborne transmission, latent infection, and drug resistance change management. Influenza produces seasonal morbidity but also destabilizes older adults, pregnant patients, immunocompromised hosts, and patients with cardiopulmonary disease. Meningitis is one of the clearest infectious disease emergencies: minutes to hours matter because early bactericidal therapy, dexamethasone in selected adults, and supportive care can prevent death or permanent neurologic injury.

The practical goal is not to memorize every organism. It is to recognize the syndrome, estimate severity, identify host-specific risks, obtain cultures when they are likely to change care, start empiric therapy when delay is dangerous, and narrow therapy once microbiology and clinical trajectory become clear.

Core illness scripts

Syndrome	Classic pattern	High-yield organisms / clues	Immediate danger
Typical community-acquired pneumonia	Acute fever, chills, productive cough, pleuritic pain, lobar infiltrate	<i>Streptococcus pneumoniae</i> ; also <i>Haemophilus influenzae</i> , <i>Moraxella</i> in COPD, <i>S. aureus</i> after influenza	Hypoxemia, sepsis, parapneumonic effusion
Atypical pneumonia	Gradual onset, headache, sore throat, dry cough, diffuse/interstitial infiltrates, extrapulmonary clues	<i>Mycoplasma</i> , <i>Chlamydia pneumoniae</i> , <i>Legionella</i> , respiratory viruses	<i>Legionella</i> can cause severe pneumonia, diarrhea, hyponatremia, renal/hepatic abnormalities
HAP/VAP	New infiltrate plus fever, purulent secretions, leukocytosis/leukopenia, worsening oxygenation after hospitalization or ventilation	Gram-negative bacilli including <i>Pseudomonas</i> , <i>Enterobacterales</i> ; <i>S. aureus</i> including MRSA in risk settings	High mortality; inappropriate empiric therapy and lack of source control worsen outcomes
Lung abscess / aspiration necrotizing infection	Indolent fever, weight loss, foul sputum, cavitary lesion often in dependent lung segments	Oral anaerobes, streptococci, mixed flora; consider <i>S. aureus</i> or gram-negative rods in severe necrotizing disease	Empyema, massive hemoptysis, treatment failure if anaerobes not covered
Tuberculosis	Chronic cough, fever, night sweats, weight loss, hemoptysis; upper lobe cavitation or miliary pattern	<i>Mycobacterium tuberculosis</i> ; risk increased with HIV, close exposure, incarceration, homelessness, immigration from high-prevalence regions	Airborne spread, delayed diagnosis, drug resistance, immune reconstitution issues
Influenza	Abrupt fever, chills, myalgias, headache, nonproductive cough, sore throat during season/outbreak	Influenza A/B; secondary bacterial pneumonia may follow	Respiratory failure, myocarditis, encephalopathy, bacterial superinfection
Acute bacterial meningitis	Fever, headache, neck stiffness, altered mental status; petechial rash suggests meningococcus	<i>S. pneumoniae</i> , <i>N. meningitidis</i> , <i>Listeria</i> in older/immunocompromised/pregnant patients	Death, seizures, hearing loss, stroke, hydrocephalus

A syndrome-first diagnostic approach

Original diagnostic flow: respiratory infection at the bedside

1. Confirm that the syndrome is lower respiratory tract disease rather than isolated bronchitis or URI: cough plus dyspnea, pleuritic pain, abnormal lung exam, hypoxemia, or infiltrate on imaging.
2. Assess severity before naming the organism: oxygen saturation, work of breathing, blood pressure, mental status, lactate if sepsis is possible, renal function, comorbidities, and ability to take oral therapy.
3. Classify setting and exposure: community onset, recent hospitalization, mechanical ventilation, aspiration risk, nursing facility exposure, structural lung disease, immunosuppression, travel, TB risk, influenza/COVID circulation.
4. Choose tests that can change management: chest radiograph for suspected pneumonia; blood and sputum cultures for severe CAP or MRSA/Pseudomonas risk; respiratory viral testing when it changes isolation or antiviral therapy; urinary Legionella/pneumococcal antigen in selected severe cases.
5. Start empiric treatment when pneumonia is likely and severity warrants therapy; do not delay antibiotics for cultures when the patient is unstable.
6. Reassess at 24-72 hours: clinical stability, fever curve, oxygen requirement, culture results, resistant organism risk, drug toxicity, and complications such as effusion, empyema, abscess, or obstructing lesion.

Community-acquired pneumonia

Community-acquired pneumonia (CAP) is usually acquired by microaspiration of organisms colonizing the upper airway. The disease becomes clinically important when pathogen burden, impaired mucociliary clearance, impaired cough, viral epithelial damage, chronic cardiopulmonary disease, altered consciousness, or immune defects allow infection to reach alveoli and produce inflammation. Alveolar exudate impairs ventilation, increases shunt physiology, and produces tachypnea and hypoxemia.

The uploaded source emphasizes the classic distinction between typical CAP, which often has abrupt fever and productive cough, and atypical CAP, which often begins more gradually with headache, sore throat, myalgias, and a dry cough. The distinction is useful for illness scripts, but real patients overlap, so empiric outpatient and inpatient regimens often cover both typical and atypical organisms.

Feature	Typical CAP	Atypical CAP	Severe or special clue
Tempo	Abrupt fever/chills, cough, pleuritic pain	Gradual constitutional symptoms, dry cough	Rapid deterioration suggests influenza-associated <i>S. aureus</i> , severe pneumococcus, Legionella, or gram-negative pneumonia
Exam	Crackles, bronchial breath sounds, dullness, egophony	Often less focal; wheezing/rhonchi may occur	Hypotension, hypoxemia, confusion, or multilobar findings imply high severity
Imaging	Lobar or segmental consolidation	Diffuse/interstitial or patchy infiltrates	Cavitation suggests <i>S. aureus</i> , anaerobic abscess, gram-negative necrotizing infection, TB, or malignancy
Organisms	<i>S. pneumoniae</i> , <i>H. influenzae</i> , <i>Moraxella</i> , <i>S. aureus</i> , <i>Enterobacterales</i>	<i>Mycoplasma</i> , <i>Chlamydia pneumoniae</i> , Legionella, respiratory viruses	Legionella: GI symptoms, hyponatremia, transaminitis, severe pneumonia; influenza: abrupt myalgia/fever
Testing	CXR usually confirms; sputum/blood cultures mainly in severe or resistant-risk patients	Viral PCR, Legionella urine antigen in severe/epidemiologic settings	Do not wait for microbiology in unstable patients

Severity triage: outpatient versus inpatient care

The most important CAP decision is not the exact organism; it is whether the patient can be safely treated as an outpatient. Objective tools such as PSI or CURB-65 support, but do not replace, bedside judgment. Red flags include oxygen saturation below the patient baseline, respiratory rate ≥ 30 /min, systolic BP < 90 mmHg, diastolic BP ≤ 60 mmHg, new confusion, multilobar infiltrates, rising BUN/creatinine, thrombocytopenia, hypothermia, lactate elevation, inability to take oral medications, poor follow-up, or decompensated comorbid disease.

Empiric CAP therapy should reflect severity and local resistance. In broad educational terms, healthy outpatients often need coverage of pneumococcus and atypicals; outpatients with comorbid disease often need broader beta-lactam plus atypical coverage or a respiratory fluoroquinolone when appropriate; inpatients commonly receive a beta-lactam plus macrolide or a respiratory fluoroquinolone; severe CAP or validated MRSA/Pseudomonas risk requires targeted additions rather than universal broad coverage.

Pearl: a radiograph can lag behind clinical improvement for weeks. Do not declare treatment failure solely because the x-ray remains abnormal if fever, oxygenation, cough, and systemic symptoms are improving. Conversely, worsening oxygenation, persistent fever, bacteremia, pleural fluid, or hemoptysis should trigger reassessment for empyema, abscess, resistant pathogen, pulmonary embolism, malignancy, or noninfectious mimic.

Hospital-acquired and ventilator-associated pneumonia

Hospital-acquired pneumonia (HAP) occurs after hospitalization and is not incubating at admission. Ventilator-associated pneumonia (VAP) occurs in mechanically ventilated patients and is promoted by impaired cough, microaspiration around the endotracheal tube cuff, biofilm formation, supine positioning, altered consciousness, and antibiotic selection pressure. These infections have higher rates of resistant gram-negative bacilli and *S. aureus* than routine outpatient CAP.

Question	HAP/VAP reasoning point
When to suspect	New or progressive infiltrate plus purulent secretions, fever/hypothermia, leukocytosis/leukopenia, worsening oxygenation, or increased ventilator requirements. No single sign is definitive.
Microbiology	Common targets are <i>Pseudomonas aeruginosa</i> and other gram-negative rods, Enterobacteriales, <i>Acinetobacter</i> in some units, and MRSA when risk factors/local prevalence are present. Local antibiograms matter.
Cultures	Respiratory sampling helps de-escalation; blood cultures may identify bacteremia or another source. Quantitative BAL is not always required, but high-quality lower respiratory samples are useful.
Empiric therapy	Use one or more antipseudomonal agents depending on severity, local resistance, prior IV antibiotics, and risk of mortality. Add MRSA coverage when risk is meaningful. Avoid unnecessary triple therapy once data return.
De-escalation	Narrow at 48-72 hours when cultures and clinical response permit. Shorter courses are often adequate if clinical improvement is clear and source control is achieved.
Prevention	Head-of-bed elevation, oral care protocols, sedation minimization, spontaneous breathing trials, subglottic secretion drainage in selected devices, hand hygiene, and avoiding unnecessary ventilation.

Original contrast: CAP versus HAP/VAP

Dimension	CAP	HAP/VAP
Typical entry point	Microaspiration in community host	Hospital airway/aspiration ecology plus device/antibiotic pressure
Most common historical anchor	<i>S. pneumoniae</i> remains central	Gram-negative rods and <i>S. aureus</i> become more prominent
Testing threshold	Outpatient microbiology often unnecessary; severe/risk-based cultures	Respiratory cultures are more useful for de-escalation
Treatment mistake	Undertreating severity or missing MRSA/ <i>Pseudomonas</i> risk	Overly broad therapy without cultures, then failing to de-escalate
Complication focus	Effusion/empyema, respiratory failure, abscess	Ventilator dependence, septic shock, resistant pathogens

Parapneumonic effusion, empyema, and lung abscess

A parapneumonic effusion is pleural fluid associated with pneumonia. Many are sterile inflammatory effusions that resolve with antibiotics. Empyema is infected pleural space and requires drainage because antibiotics alone penetrate poorly into loculated purulent collections. A lung abscess is a necrotic cavity within lung parenchyma, classically after aspiration of oral anaerobes into dependent segments. The source pages emphasize that abscesses are usually larger cavitory lesions and that therapy may require prolonged antibiotics until the cavity improves markedly or resolves.

Finding	Uncomplicated parapneumonic effusion	Complicated effusion / empyema	Lung abscess
Location	Pleural space	Pleural space, often loculated	Lung parenchyma cavity
Clues	Small free-flowing effusion, clinical improvement	Persistent fever, pleuritic pain, large/loculated effusion, frank pus, low pH/glucose	Foul sputum, aspiration risk, air-fluid level, thick-walled cavity
Testing	Often no thoracentesis if tiny and improving	Thoracentesis: Gram stain/culture, pH, glucose, LDH, cell count; ultrasound/CT for loculations	CXR/CT; consider bronchoscopy if obstruction, unusual organism, or failure
Management	Treat pneumonia	Antibiotics plus drainage; consider chest tube, fibrinolytics, or surgery if needed	Anaerobic and oral flora coverage; prolonged therapy; drainage rarely unless large/refractory or complication

Aspiration does not equal anaerobes in every patient. Brief witnessed aspiration pneumonitis may be chemical and can improve quickly without antibiotics. Persistent fever, leukocytosis, infiltrates beyond 48 hours, periodontal disease,

necrotizing disease, abscess, or empyema should raise concern for bacterial aspiration pneumonia with anaerobic/oral flora involvement.

Tuberculosis

Tuberculosis is caused by *Mycobacterium tuberculosis*, an acid-fast, slow-growing organism transmitted by airborne droplet nuclei from patients with active pulmonary or laryngeal disease. Primary infection begins when inhaled bacilli reach alveoli, survive macrophage ingestion, and provoke a delayed cellular immune response. Granulomas contain infection but do not always sterilize it, which is why latent infection can reactivate years later when immunity wanes.

TB reasoning requires separating exposure, latent infection, active pulmonary disease, active extrapulmonary disease, and drug resistance. A positive screening test supports infection but does not prove active disease. Active disease is diagnosed by compatible syndrome plus imaging, acid-fast smear, nucleic acid amplification testing, culture, and susceptibility testing. Culture remains important because treatment duration and drug-resistance management depend on the organism and its susceptibilities.

Concept	Key distinctions
Latent TB infection	Positive IGRA or TST with no symptoms and no radiographic/microbiologic evidence of active disease. Not contagious. Treat to reduce future reactivation risk after active disease is excluded.
Active pulmonary TB	Cough often >2-3 weeks, fevers, night sweats, weight loss, hemoptysis, apical infiltrates/cavitation or atypical findings in HIV/early disease. Airborne isolation is required.
Extrapulmonary TB	Lymph node, pleural, genitourinary, bone/spine, meningitis, peritoneal, adrenal, or miliary disease. Pulmonary symptoms may be absent.
Risk groups	Close contacts, prior untreated TB, birth/residence in high-prevalence region, homelessness, incarceration, HIV, transplant, TNF-alpha inhibitors, silicosis, diabetes, chronic kidney disease, substance use.
Testing	IGRA/TST for infection; CXR for active pulmonary suspicion; sputum AFB smear/culture and NAAT for active disease. Smear positivity correlates with infectiousness but smear-negative TB can occur.
Treatment frame	Drug-susceptible active TB is treated with multidrug therapy using an intensive phase followed by a continuation phase. Latent TB has shorter rifamycin-based options and is never treated until active disease is excluded.

Original TB algorithm: from suspicion to public health action

1. Identify risk plus syndrome: chronic cough, fever, night sweats, weight loss, hemoptysis, abnormal CXR, TB exposure, or immunocompromise.
2. If active pulmonary TB is possible, place airborne isolation immediately and notify infection control/public health according to local process.
3. Obtain chest imaging and respiratory specimens for AFB smear, mycobacterial culture, NAAT, and drug-susceptibility testing. Do not rely on TST/IGRA to rule out active disease.
4. If active TB is likely or the patient is ill, start multidrug therapy after specimens are collected when feasible; add pyridoxine with isoniazid in patients at neuropathy risk and in common practice.
5. Monitor hepatotoxicity, drug interactions, adherence, microbiologic response, and contact investigation. Directly observed therapy or other adherence supports are often used.
6. For latent TB, exclude active disease first; then choose a regimen based on drug interactions, pregnancy status, liver risk, exposure source susceptibility, and ability to complete therapy.

High-yield TB traps

- BCG vaccination can cause false-positive tuberculin skin tests, but IGRA is more specific in BCG-vaccinated people; prior BCG does not change the need to evaluate possible active TB.
- HIV and severe immunosuppression can produce atypical chest imaging, negative sputum smears, extrapulmonary disease, and false-negative screening tests.
- Rifampin/rifapentine are potent drug interaction medications; they can reduce effectiveness of many drugs including warfarin and some hormonal contraceptives.
- Isoniazid can cause hepatitis and peripheral neuropathy; pyridoxine reduces neuropathy risk but does not prevent hepatitis.
- Never use single-drug therapy for active TB because resistance can emerge.

Influenza

Influenza is an orthomyxovirus transmitted by respiratory droplets and aerosols, with seasonal outbreaks driven by antigenic drift. Abrupt fever, chills, myalgias, headache, sore throat, and nonproductive cough are classic. The disease is often self-limited in healthy adults but can cause severe pneumonia, secondary bacterial pneumonia, myocarditis, encephalopathy, rhabdomyolysis, and decompensation of asthma, COPD, heart failure, or frailty.

Clinical decision	Medvale anchor
When to test	During influenza circulation, test hospitalized patients, high-risk patients, outbreak settings, and when results alter antiviral/isolation decisions. A negative rapid antigen test is less reliable than molecular testing.
Who needs antivirals	Hospitalized, severe/progressive disease, high-risk patients, pregnancy/postpartum, older adults, and many patients within 48 hours of symptom onset. Do not withhold treatment from hospitalized or severe cases just because >48 hours have passed.
Drug class	Neuraminidase inhibitors such as oseltamivir are commonly used. Baloxavir is an option in selected outpatient settings but is not the universal default.
When to suspect bacterial superinfection	Initial improvement followed by recurrent fever, productive cough, focal consolidation, shock, leukocytosis, or severe necrotizing pneumonia. Consider <i>S. aureus</i> and <i>S. pneumoniae</i> .
Prevention	Annual vaccination, respiratory hygiene, isolation precautions in healthcare, post-exposure prophylaxis in selected high-risk settings.

Meningitis and CNS infections

Meningitis is inflammation of the meninges. Acute bacterial meningitis is a medical emergency because organisms in the subarachnoid space trigger cytokine-mediated inflammation, blood-brain barrier disruption, cerebral edema, vasculitis, impaired CSF flow, elevated intracranial pressure, seizures, and infarction. The classic triad is fever, neck stiffness, and altered mental status, but many adults do not present with all three. Headache, photophobia, nausea/vomiting, rash, seizure, and focal neurologic deficits can occur.

The source section highlights that CSF evaluation is central but that treatment should not wait when bacterial meningitis is likely. The sequencing rule is practical: obtain blood cultures and give antibiotics promptly; perform lumbar puncture immediately unless there is a reason for CT first; if CT is needed before LP, antibiotics and dexamethasone when indicated should be given before imaging rather than delayed.

CSF pattern	Opening pressure	WBC count/differential	Glucose	Protein	Typical interpretation
Normal	Normal	0-5 cells/uL, mostly lymphocytes/monocytes	Normal	Normal	No meningitis, though early disease can be subtle
Bacterial meningitis	Often high	High, often 1,000-5,000/uL; neutrophils predominate	Low relative to serum	High	<i>S. pneumoniae</i> , <i>N. meningitidis</i> , <i>Listeria</i> , <i>H. influenzae</i> , gram-negative bacilli depending on age/risk
Viral/aseptic meningitis	Normal or mildly high	Lower count; lymphocytes predominate after early phase	Usually normal	Normal to moderately high	Enterovirus, HSV, VZV, HIV, arboviruses; clinical severity varies
TB/fungal meningitis	Often high	Lymphocytic predominance	Low	High	Subacute/chronic course, cranial neuropathies, basilar meningitis, hydrocephalus; requires special cultures/tests
Partially treated bacterial	Variable	May be lower; neutrophils may persist	May be low	High	Prior antibiotics can reduce culture yield; PCR and clinical context matter

When to image before lumbar puncture

CT before LP is considered when there are signs suggesting mass effect or dangerous intracranial pressure shift: immunocompromised state, history of CNS disease, new seizure, papilledema, abnormal level of consciousness, or focal neurologic deficit. If CT is needed, draw blood cultures and start empiric antibiotics immediately; do not make antibiotics wait for CT or LP.

Age/risk group	Empiric coverage concept
Young healthy adult	Vancomycin plus a third-generation cephalosporin is a common core to cover resistant pneumococcus and meningococcus.
Age >50, pregnancy, alcoholism, immunocompromised, impaired cellular immunity	Add ampicillin for <i>Listeria</i> coverage.
Post-neurosurgery, penetrating trauma, shunt, healthcare-associated ventriculitis/meningitis	Cover resistant staphylococci and gram-negative bacilli including <i>Pseudomonas</i> ; regimen depends on local guidance and hardware status.
Concern for HSV encephalitis	Add acyclovir, especially with temporal lobe findings, seizures, focal deficits, or marked altered mental status.
Meningococcal disease	Droplet precautions; chemoprophylaxis for close contacts according to public health guidance.

Original meningitis emergency algorithm

1. Recognize possible meningitis: fever, severe headache, neck stiffness, altered mental status, petechiae/purpura, seizure, photophobia, or toxic appearance.

2. Stabilize airway, breathing, circulation. Check glucose because hypoglycemia can mimic altered mental status.
3. Obtain blood cultures immediately if this does not cause major delay.
4. If no CT-before-LP criteria, perform LP promptly and send CSF cell count/differential, glucose with serum glucose, protein, Gram stain/culture, and targeted PCR/antigen tests.
5. If CT is needed before LP or LP will be delayed, give empiric antimicrobials immediately after blood cultures. Add dexamethasone before or with first antibiotic dose in appropriate suspected pneumococcal meningitis settings.
6. Refine therapy using Gram stain, culture, PCR, susceptibility, age/risk profile, and clinical course. Manage complications: seizures, hydrocephalus, shock, hyponatremia, hearing loss, and focal neurologic deficits.

Integrated differential: infection versus mimic

Presentation	Infectious possibilities	Important mimics / alternate diagnoses	Trap
Fever + cough + infiltrate	CAP, aspiration pneumonia, influenza, TB, fungal pneumonia, opportunistic pneumonia	Pulmonary embolism, pulmonary edema, malignancy, organizing pneumonia, vasculitis	Do not anchor on pneumonia if hypoxemia, pleuritic pain, and risk factors strongly suggest PE.
Cavitary lung lesion	TB, lung abscess, S. aureus, Klebsiella, anaerobes, fungal infection, septic emboli	Squamous cell carcinoma, granulomatosis with polyangiitis, rheumatoid nodules	Abscess and empyema can look similar; CT helps determine parenchymal versus pleural location.
Subacute fever + night sweats + weight loss	TB, HIV-related infection, endocarditis, lymphoma-associated infection	Lymphoma, autoimmune disease, malignancy	A negative initial smear does not fully exclude TB.
Fever + headache + neck stiffness	Bacterial meningitis, viral meningitis, HSV/VZV, TB/fungal meningitis	Subarachnoid hemorrhage, migraine, encephalitis, drug fever	Do not delay antibiotics for imaging if LP is delayed.
Influenza-like illness with worsening after improvement	Influenza with secondary bacterial pneumonia	Heart failure, asthma/COPD flare, COVID or other viral infection	Post-influenza S. aureus pneumonia can be severe and necrotizing.

Initial management bundles

CAP initial bundle

- Assess oxygenation, blood pressure, respiratory rate, mental status, and comorbidity burden.
- Obtain chest imaging when pneumonia is suspected; do not treat isolated uncomplicated bronchitis with antibiotics.
- For admitted/severe patients, obtain appropriate cultures before antibiotics if this does not delay treatment.
- Start empiric therapy based on severity, comorbidities, MRSA/Pseudomonas risk, aspiration/abscess concern, and local resistance.
- Reassess at 48-72 hours for stability, de-escalation, oral switch, complications, or alternate diagnosis.

TB initial bundle

- Airborne isolation for suspected active pulmonary/laryngeal TB.
- Collect sputum for AFB smear, mycobacterial culture, NAAT, and susceptibilities.
- Coordinate public health notification, contact tracing, adherence planning, and discharge isolation criteria.
- Use multidrug therapy for active disease; never monotherapy.
- Check baseline liver risk and drug interactions; counsel on orange body fluids with rifamycins and neuropathy symptoms with isoniazid.

Meningitis initial bundle

- Give empiric antibiotics urgently; obtain blood cultures first when feasible.
- Perform LP unless CT-first criteria are present; if CT delays LP, treat first.
- Add age/risk-specific coverage such as ampicillin for Listeria risk or acyclovir for encephalitis concern.
- Use droplet precautions for suspected meningococcal disease and notify public health for prophylaxis of close contacts.
- Monitor for seizures, increased intracranial pressure, shock, respiratory failure, hearing loss, and hyponatremia.

High-yield exam clues and bedside anchors

Clue	Diagnosis suggested	Why it matters
Rust-colored sputum, lobar consolidation	Pneumococcal pneumonia	Classic but not required; can cause bacteremia and severe CAP
Alcohol use disorder + currant jelly sputum + bulging fissure	Klebsiella pneumonia	Necrotizing gram-negative pneumonia; aspiration risk often overlaps
Bird exposure	Chlamydia psittaci	Atypical pneumonia exposure history
Farm animals/parturient animals	Coxiella burnetii (Q fever)	Atypical pneumonia plus hepatitis/endocarditis associations
Hyponatremia + diarrhea + severe pneumonia	Legionella	Urine antigen may help for L. pneumophila serogroup 1
Foul sputum + aspiration risk + air-fluid level	Lung abscess	Anaerobic oral flora coverage and prolonged therapy
Upper-lobe cavitation + night sweats + weight loss	Reactivation TB	Airborne isolation, AFB/NAAT/culture, public health
Abrupt myalgias + fever in winter outbreak	Influenza	Early antivirals for high-risk/severe disease; watch bacterial superinfection
Fever + petechiae/purpura + meningismus	Meningococcemia/meningitis	Droplet precautions, urgent antibiotics, contact prophylaxis
CSF neutrophils + low glucose + high protein	Bacterial meningitis	Immediate antibiotics; tailor once cultures/PCR return

Common pitfalls

- Treating every cough as bacterial pneumonia. Pneumonia requires compatible syndrome and usually an infiltrate; acute bronchitis is most often viral.
- Ignoring oxygen saturation and work of breathing. A patient with a small infiltrate but hypoxemia may be sicker than a patient with a dramatic x-ray and normal physiology.
- Using broad HAP/VAP therapy without obtaining cultures or without de-escalation. Broad therapy is a bridge, not a destination.
- Missing empyema because the pneumonia was treated. Persistent fever or pleural fluid requires reassessment and sometimes drainage.
- Relying on a negative TB smear or a negative screening test to exclude active TB in high-risk or immunocompromised patients.
- Treating latent TB before excluding active TB. Monotherapy in active disease promotes resistance.
- Delaying bacterial meningitis antibiotics for CT, LP, transfer, or paperwork. Blood cultures and immediate therapy are the priority when LP is delayed.
- Forgetting Listeria coverage in older, pregnant, alcoholic, or immunocompromised patients with suspected bacterial meningitis.
- Stopping pneumonia antibiotics solely because the chest x-ray has not cleared. Radiographic resolution often lags clinical recovery.

Practice questions

Question 1. A 62-year-old with COPD has fever, productive cough, pleuritic pain, oxygen saturation 89%, and right lower lobe consolidation. What are the first two decisions?

Answer: First assess severity and site of care; second choose empiric therapy based on CAP severity/comorbidities and local resistance. Oxygenation and comorbid COPD make outpatient management less automatic.

Question 2. A mechanically ventilated patient develops fever, purulent secretions, increased oxygen requirement, and new infiltrate on day 6 of admission. What organisms must empiric therapy consider?

Answer: VAP should consider gram-negative bacilli including *Pseudomonas* and *S. aureus* including MRSA if risk factors/local prevalence support it. Respiratory cultures should guide de-escalation.

Question 3. A man with alcohol use disorder has fever, foul sputum, weight loss, and a cavitary lesion with air-fluid level in a dependent lung segment. What is the key treatment concept?

Answer: Lung abscess from aspiration requires anaerobic/oral flora coverage and prolonged therapy. Evaluate for obstruction, empyema, or unusual pathogens if the course is atypical.

Question 4. A patient from a high-TB-prevalence country has chronic cough, night sweats, weight loss, and apical cavitation. What should happen before routine ward placement?

Answer: Airborne isolation and respiratory specimens for AFB smear, mycobacterial culture, NAAT, and susceptibility testing. Public health/infection control involvement is necessary.

Question 5. An older patient presents with fever, confusion, neck stiffness, and new seizure. CT is ordered before LP. What should not wait?

Answer: Blood cultures and empiric meningitis therapy should not wait for CT or LP. Add ampicillin for *Listeria* risk and consider acyclovir if encephalitis is possible.

Question 6. CSF shows 2,500 WBC/uL with neutrophil predominance, low glucose, and high protein. What pattern is this?

Answer: Bacterial meningitis until proven otherwise. Treat immediately and narrow using Gram stain, culture, PCR, and risk profile.

Question 7. A hospitalized influenza patient initially improves but then has recurrent fever, leukocytosis, shock, and new cavitary pneumonia. What complication is suggested?

Answer: Secondary bacterial pneumonia, including severe *S. aureus* necrotizing pneumonia, should be considered.

Graph-RAG extraction map

The following entities and relationships are designed for later conversion into concept nodes, edge classes, and source-linked retrieval snippets.

Node	Suggested classes	High-value edges
Community-acquired pneumonia	infectious_syndrome; pulmonary_infection; acute_care	caused_by S. pneumoniae / atypicals / viruses; diagnosed_by CXR; treated_by empiric_CAP_antibiotics; complicated_by effusion/empyema/respiratory_failure
Atypical pneumonia	pulmonary_infection; microbiology_pattern	associated_with dry_cough/interstitial_infiltrate; caused_by Mycoplasma/Chlamydia/Legionella; treated_by macrolide/doxycycline/respiratory_fluoroquinolone depending context
Legionella pneumonia	atypical_pneumonia; gram_negative; water_exposure	associated_with hyponatremia/diarrhea/transaminitis; diagnosed_by urine_antigen/PCR/culture; treated_by macrolide_or_fluoroquinolone
HAP/VAP	healthcare_associated_infection; pulmonary_infection; device_associated	risk_increased_by ventilation/prior_antibiotics; caused_by Pseudomonas/MRSA/gram_negatives; requires cultures; managed_by broad_empiric_then_deescalation
Lung abscess	suppurative_infection; aspiration_related	caused_by oral_anaerobes; diagnosed_by cavitary_air_fluid_level; treated_by prolonged_anaerobic_coverage; differentiated_from empyema
Tuberculosis	airborne_infection; mycobacterial_disease; public_health	caused_by M_tuberculosis; transmitted_by airborne_droplets; diagnosed_by AFB_culture_NAAT; treated_by multidrug_regimen; prevented_by latent_TB_treatment
Latent TB infection	infection_state; preventive_medicine	screened_by IGRA/TST; must_exclude active_TB_before_treatment; progresses_to active_TB_when_immunity_wanes
Influenza	viral_respiratory_infection; seasonal_infection	caused_by influenza_A/B; prevented_by vaccination; treated_by neuraminidase_inhibitor_in_high_risk; complicated_by bacterial_pneumonia
Bacterial meningitis	CNS_infection; emergency	caused_by pneumococcus/meningococcus/Listeria; diagnosed_by CSF; treated_by urgent_antibiotics; complicated_by seizures/hearing_loss/death
CSF profile	diagnostic_test_pattern; laboratory_reasoning	distinguishes bacterial_vs_viral_vs_TB_fungal; includes WBC_differential/glucose/protein/opening_pressure
Empiric antibiotic therapy	therapeutic_strategy; stewardship	selected_by syndrome/severity/host_risk/local_resistance; narrowed_by culture_response; risk_of_toxicity/resistance

References used for clinical cross-checking

American Thoracic Society and Infectious Diseases Society of America. Diagnosis and treatment of adults with community-acquired pneumonia. 2019 ATS/IDSA guideline.

Infectious Diseases Society of America and American Thoracic Society. Management of adults with hospital-acquired and ventilator-associated pneumonia. 2016 guideline.

Centers for Disease Control and Prevention. Tuberculosis clinical care, diagnosis, infection control, and treatment resources.

Infectious Diseases Society of America. Practice guidelines for the management of bacterial meningitis and healthcare-associated ventriculitis/meningitis resources.

Centers for Disease Control and Prevention. Influenza antiviral medications and seasonal influenza clinical guidance.

Uploaded source section: infection1.pdf, used as a topic coverage guide for pneumonia, VAP, lung abscess, TB, influenza, and meningitis.